

# Runbao (Frank) Du

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## EDUCATION

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### University of Chicago, The College — Chicago, IL

Aug 2025 – May 2027 (expected)

*Bachelor of Science in Computational and Applied Mathematics; Minor in Astronomy and Astrophysics*

- Cumulative GPA: 3.81/4.00
- Advanced coursework: Abstract Linear Algebra; Analysis in  $\mathbb{R}^n$  I–III; Mathematical Foundations of Machine Learning; Introduction to Astrophysics; Black Holes; Stars

### University of Southern California — Los Angeles, CA

Aug 2023 – May 2025

*Bachelor of Science in Applied and Computational Mathematics; Minor in Philosophy (transferred to the University of Chicago)*

- Cumulative GPA: 4.00/4.00
- Advanced coursework: Theory of Numbers; Probability Theory; Seminar in Problem Solving; Discrete Methods in Computer Science; Theory of Knowledge; Human Language and Technology; Creating the Motion Picture Sound Track
- Informal Reading Session on Number Theory with Prof. Christian Geske

### Additional Coursework

- *Cryptography I*, Stanford University Online — certificate of completion

## HONORS AND AWARDS

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- **2026 AIx Origin Hackathon, ModelScope Open-Source Contribution Award** - 1/70 Teams.
- **2026 Shanghai WaiTan Hackathon, BAIDU Track Award, REDNOTE Track Award** - Top 1.1% (BAIDU Track), Top 100 (REDNOTE Track).
- **USC Dornsife Dean's List** — four times, 2023–2025.
- **Duolingo English Test Merit Scholarship** — **sole recipient** worldwide, CNY 10,000 award. Selection comprised an examination, written responses, and a public address delivered to fellow test takers.
- **Student Investment Challenge, Season 6 Global Finals** — Individual **Gold** in Financial Theory (top three overall); Silver, Best Overall Team; Silver, Best Strategy Showcase; Bronze, Best Investment Report. Returned in Season 7 as Peer Mentor to five teams and served as the inaugural guest of the organization's official *FinTalk* series.
- **2022-23 C-IDEA Design Award**, Visual Category — 6.8% average acceptance rate. Submission of 10 posters and 7 mockups on environmental themes (Photoshop, Illustrator, Affinity).
- **Poetry published in *NYT for Kids CHINA***, December 2022 issue — “A Long Time,” a bilingual poem in Japanese verse form.
- **Permanent member, Music Copyright Society of China (MCSC)** — contributing royalty-free catalog for commercial use and promoting copyright knowledge across music communities.

## RESEARCH EXPERIENCE

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## Research Lead — A Statistical and Philosophical Analysis of the “Humanness” in Human Music

May 2026 – present

*University of Chicago, Department of Computer Science. Advisor: Prof. Blase Ur*

- This study explores what makes human music sound human, and approaches the question from two independent sources of evidence.
- Measure interpretable audio features of AI-generated and human-made music through a pipeline of audio analysis techniques. At the same time, the study also includes listening studies involving different groups of listeners, from the general public to musicians and recording engineers.
- The study then compares the two sets of findings to identify which aspects of humanness are both measurable and audible, which only listeners perceive, and which only the statistics detect.

## Researcher — Problems in deepfake detection in AI-generated music and the construction of incremental model as a solution

Feb 2026 –

present

*Fudan University. Advisor: Qile He*

- Developing an enrollment-based incremental detector for AI-generated music, and testing whether it outperforms general-purpose detectors such as SpecTtTra under matched data and compute budgets.
- Built an audited dataset of about 98,600 tracks (43,400 human tracks and 55,200 AI-generated tracks from 12 generators across 7 architecture families), with provenance and contamination audits on both the human and the AI side.
- The study so far has found that each generator’s deviation direction in frozen self-supervised feature space clusters by architecture family, and that pairwise alignment predicts cross-generator transfer (Spearman  $\rho$  up to  $-0.86$ ).

## Researcher — The Effects of Value-Laden Persona Descriptions on LLM Value Inferences and Task Completion

Nov 2025 – present

*University of Chicago, The SUPERGroup Lab.*

- Experiment I: measured the values that eight large language models infer from 33 persona descriptions, yielding 2,583 distinct values organized into 162 clusters; only 99 values (3.83%) were shared by all eight models.
- Experiment II: measured how those personas shift model behavior across 33 downstream tasks. Inferred value profiles were near-invariant under resampling of the same primer (mean cosine 0.984) but shifted sharply under paraphrase of the same value (0.558).
- Responsible for prompt engineering, output schema design, methodology designing, and writing a considerable portion of the manuscript.

## Researcher — MuseV2, Controllable Text-to-Music Generation

June 2026 - present

*Fudan University Natural Language Processing Laboratory*

- One of the **four initial members**; designed a deterministic cascading DAG prompt generator for AI-generated music data, covering 17 genre families and 99 sub-genres that emits structured song specifications with balanced field coverage and principled conditional dependence between fields.
- Each field inherits hierarchically from a genre-family profile and is composed through traits and three latent musical axes (aggression, density, acousticalness); because generation proceeds from anchors by inference rather than by random draw from a descriptor pool, tag conflicts are structurally impossible rather than patched after the fact.

- Authored the base configuration for each genre personally, drawing on professional production experience to specify what each style should sound like.
- Also designing the complementary pipeline on the human side: extraction and tagging of real recordings into the same structured schema the prompt generator emits, so that model-generated specifications and human training data are described in one vocabulary rather than two. Serves as the human intervention in that loop, continuously auditing tag output against the audio and feeding corrections back into the tagger.

### Mentee — Directed Reading Program

Oct 2025 – Jan 2026

*University of Chicago, Department of Mathematics*

- Studied algebraic geometry and its application to elliptic curves from Silverman’s *The Arithmetic of Elliptic Curves*, working toward the proof of the elliptic curve primality test.
- Covered affine and projective varieties, Hilbert’s Nullstellensatz, smoothness, the short Weierstrass form, and the group law on  $E(k)$ , as a precursor to morphisms of curves, Hasse’s theorem, and the Goldwasser–Kilian algorithm.
- Delivered a public talk, *A Premise to the Elliptic Curve Primality Test*, on 15 January 2026.

### Researcher — Segmentation of phrases in the album *Donda* by Kanye West and their effects and purpose

2022 - 2023

*Oberlin College. Advisor: Prof. Jan Miyake*

- Wrote a 27-page research paper analyzing phrase segmentation across all 27 tracks of Kanye West’s *Donda*, applying the Duinker–Adams two-layer framework (flow layer: rhyme, syntax, breath; beat layer: melody, harmony, meter) and identifying a progressive shift from aligned to partially aligned layers across the album.
- Paper: A– (36/40); course grade A (96/100); top 5% of the Pioneer Academics Spring 2022 cohort. The paper was subsequently presented by Prof. Miyake in her own teaching as a model for combining contemporary genre with formal music-theoretic analysis.

## PUBLICATIONS AND PRESENTATIONS

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### Manuscripts under review

- A. Row, M. Pickering, **R. Du**, A. Haniph, A. Arunasalam, and B. Ur. “The Effects of Value-Laden Persona Descriptions on LLM Value Inferences and Task Completion.” In revision for submission to *ACL Rolling Review*, October 2026 cycle.

### Talks

- **R. Du**. “A Premise to the Elliptic Curve Primality Test.” Directed Reading Program, Department of Mathematics, University of Chicago, 15 January 2026.

### Books

- **R. Du**. *Following Time*. Silicon Times, 2022. A book on self-centered time management and a three-step planning method, recognized by the China Writers Association.

## PROJECTS/BUILD

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### brain2 — Constructing, maintaining, and "Agentalizing" knowledge

Sep 2026 – present

*Founder. [ModelScope Demo Link](#)*

- brain2 helps a user quickly build, organize, and visualize personal knowledge bases, which we call second brains: notes can be directly typed, recorded, or imported from local files, split and embedded into MongoDB, clustered by topic, and laid out on a 3D map.

- Each second brain is then represented by an agent, and a four-role debate engine with mid-similarity cross-brain retrieval lets these agents exchange ideas and produce reasoned, source-traceable conclusions. The long-term goal is to connect many people's second-brain groups into a network that can be embedded in companies and teams, so that skills and knowledge flow without barriers and collaboration becomes far more efficient.
- We built the first version at AIx Origin Hack 2026, a hackathon co-organized by universities in Hong Kong, companies, and accelerators at ShenZhen. This competition is held in person in Shenzhen. I served as team lead, originated the project idea, and owned the frontend and data layers of a three-person team.
- Out of more than 70 teams, ours was among the 17 selected for the final live pitch and received the ModelScope Open-Source Contribution Award, a single special prize given outside the top three places to the project judged both open-source and of practical social value.

### **Lyra Studio — an audio-to-image generator**

Feb 2026 – present

*Founder. [Demo Link](#) - [Please contact me for the invite code](#)*

- Built and deployed Lyra Studio, a full-stack web platform that turns a music track into images and section-by-section visual scenes.
- Developed a Python audio-analysis pipeline that measures rhythm, pitch, timbre, dynamics and song structure, and matches each upload against songs in the Lyra database and their text prompts.
- Integrated Claude as an art director that writes image briefs from these measurements and Gemini as the image model, with eight visual styles and image variations. Built the system with FastAPI, React and SQLite, with user accounts, invite-only access and per-user usage limits, and deployed it on Render.

### **Audit Copilot — LLM Agent for Multi-track Audio Dataset QC**

May 2026 – July 2026

*Contributor*

- Built for FD-LAMT (Fudan Laboratory of Audio and Music Technology).
- Internal LLM agent that converts a multi-hour-per-song manual acceptance review (file naming, directory structure, WAV format, duration alignment, arrangement checks, and many more attributes) into a 17-state machine: the agent executes objective checks and blocks for human judgment where the criterion is subjective. Architecture: Electron desktop client with a Python FastAPI sidecar, agent running in the Electron main process and invoking tools over MCP.
- Redesigned the 17-step state-machine SOP for the review phase: merged three redundant listening-quality checks, decoupled two states from a mandatory mixer window in favor of the reviewer's own tools, replaced binary responses with a **pass** / **minor issue** / **fail** template, and added hard invariants preventing cross-state contamination of LLM context.
- Reduced the agent's prompt specification from 40,674 to 36,893 characters (**−9.3%**) across two staged commits, saving approximately **2,200 input tokens per LLM call** as measured against real prompt-cache hit logs; refactored duplicated invariants into single-source references.
- Did a lot of debugging across the Electron main/renderer boundary and the Python sidecar; iterated the human-decision card UI and the MIDI piano-roll viewer.

### **Prezuricle — Multimodal AI Marketing Platform**

Jan 2024 – May 2024

*Co-founder. [prezuricle.com](#)*

- Co-founded a platform aggregating multiple large-model APIs for marketing workflows: content generation, text-to-speech, text-to-image, and voice cloning. Grew out of an earlier conversational-agent prototype.
- Owned the text-audio and text-image conversion surfaces. Traffic peaked at roughly 5,000 users

per month and now sits at 100–200 with no active development. This is the earliest applied AI product work on this CV, predating the current wave of tooling.

### **Shiyu — A Visualization Platform for Literary Works**

July 2026 - Aug 2026

*Builder*

- A WebGL poetry universe in which themes are stars and poems are orbiting exoplanets, letting readers explore, curate and publish work inside a navigable 3D galaxy.
- Its central design challenge was fusing three normally separate disciplines into one coherent product: astrophysically plausible celestial simulation, the reading rhythm of poetry, and a visual aesthetic that had to satisfy both scientific credibility and literary feeling.

## **TEACHING EXPERIENCE**

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### **Grader, Department of Mathematics — University of Chicago**

Sep 2025 – May 2026

- Graded problem sets and proofs for Calculus I, Calculus II, and Abstract Linear Algebra.

### **Teaching Assistant, Keystone Academy Summer Program**

July 2023 – Aug 2023

- Lectured, designed classroom activities pairing board games with environmental content, and led a soil-pH laboratory session; additionally responsible for the safety and care of students with special needs. Organized after-school athletics and weekend programming. Compensated.

## **LEADERSHIP AND ACTIVITIES**

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### **Vice President — Southern California Math Competition, Los Angeles, CA**

Feb 2024 – May 2025

- Co-led the only student-initiated mathematics competition in Southern California recognized by the university, hosted under a `usc.edu` domain.
- Secured over \$1,000 in funding through university organizations and external fundraising; ran one competition per term drawing **150+ participants** each from surrounding high schools, **400+ students** in total.
- Managed branch operations through Notion and Slack. Past participants have gone on to further distinctions, including an International Mathematical Olympiad gold medal.

### **Campus Partner — Midea Group**

Aug 2024 – May 2025

- Served as liaison between Midea and USC student organizations, coordinating resources for the company's events in Los Angeles at roughly five hours per week.
- Delivered a recruiting information session with **150+ attendees** through joint social and on-campus promotion; promoted to lead the campus partners of the general western region on the strength of sustained delivery and candidate referrals, and held a top-five position on the Campus Partner program's standings throughout.

### **Advisory Board Member and Social Media Leader — Pioneer Academics**

2022 – May 2023

- Invited to the Advisory Board following the Spring 2022 research cohort. Managed the program's social accounts, designed its Instagram templates, and maintained the records connecting alumni with prospective and current students.

### **Founder and President — Keystone Model United Nations**

High school

- Designed a curriculum covering the full MUN process from scratch and led delegations to NISMUM. Grown to 100+ members and alumni and recognized as one of the few core societies among more than thirty at the school.

#### Member — Math Clubs

Aug 2023 – Present

- USC math club and UChicago math club. Weekly problem sessions across topology, number theory, algebra, and probability. Participated in Putnam and earned a **shocking** 2/120.

## CREATIVE WORK

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#### Recording Artist and Producer — “Frاند”

*Primary platform: NetEase Cloud Music*

- **13,000,000+ cumulative streams** and 500K streaming hours.
- **Joined the NetEase Cloud Village User Committee.**
- Debut album scheduled for May 2027.

#### Poet

- Publishes at [runbaos.com](https://runbaos.com); **200,000+ likes** on Douyin.

#### Visual Designer

2022 – 2024

- Poster design and 3D Modeling; Design portfolio can be viewed on my personal website; recipient of the C-IDEA Design Award (see Honors).

## EMPLOYMENT AND INTERNSHIPS

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#### Intern — China CITIC Bank, Beijing

May 2024 – June 2024

- Supported wealth managers in responding to routine client inquiries at a retail branch; additionally assisted non-native Mandarin speakers due to the bank’s proximity to Tsinghua University.

*See also:* Grader, University of Chicago (Teaching Experience); Campus Partner, Midea Group (Leadership and Activities); Teaching Assistant, Keystone Academy (Teaching Experience).

## INTERNATIONAL EXPERIENCE AND LANGUAGES

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- **Languages:** Mandarin Chinese (native); English (full professional proficiency). TOEFL 115 (2022); Duolingo English Test 150 (2023); SAT 1520 (2022).
- **Education across two countries:** secondary education in China; undergraduate study in the United States at USC and the University of Chicago. Ongoing research collaborations with two laboratories at Fudan University and a laboratory at the University of Chicago.

## TECHNICAL SKILLS

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**Programming and markup:** Python, C++, L<sup>A</sup>T<sub>E</sub>X, Git

**AI-centric Tools:** Claude Code and Cowork, Cursor, Obsidian, Suno, ACE Studio, Runpod, Perplexity - Comet

**Machine learning and audio:** MIR feature-extraction pipelines (MFCC, chroma, spectral and dynamic descriptors), prompt and loop engineering, LLM tool use and the Model Context Protocol, Agentic Skills writing

**Music Production and Design:** FL Studio, Logic Pro, Piano (ABRSM LEVEL 8), Adobe Photoshop, Affinity Photo, Krea

